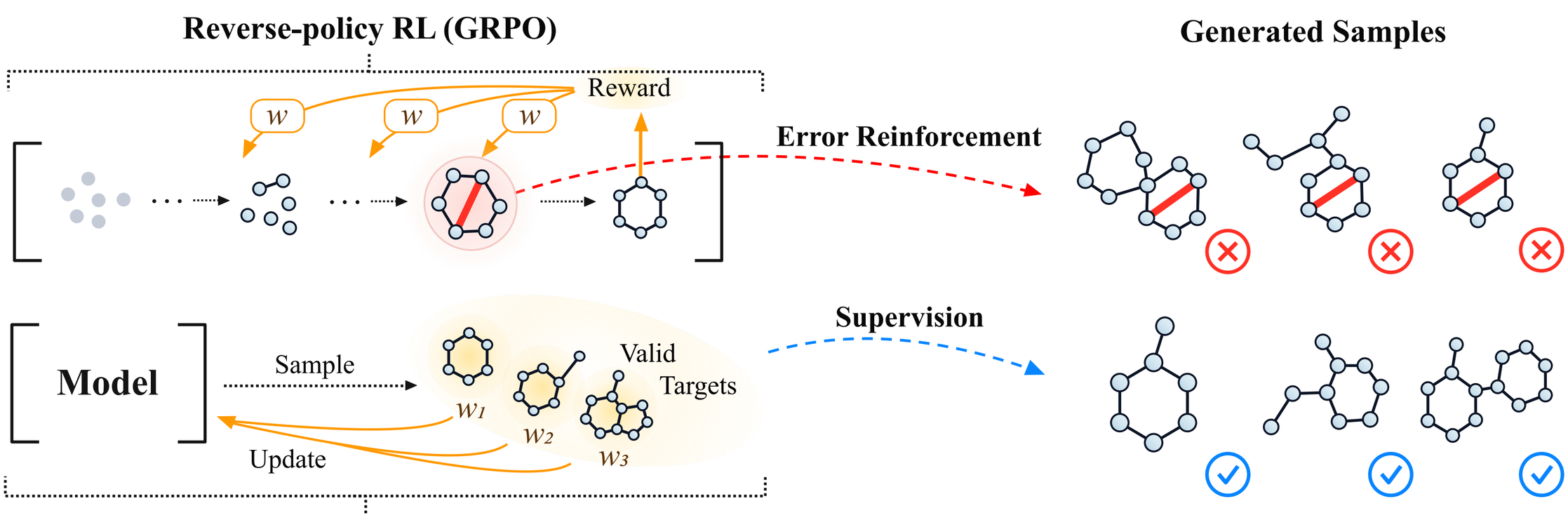
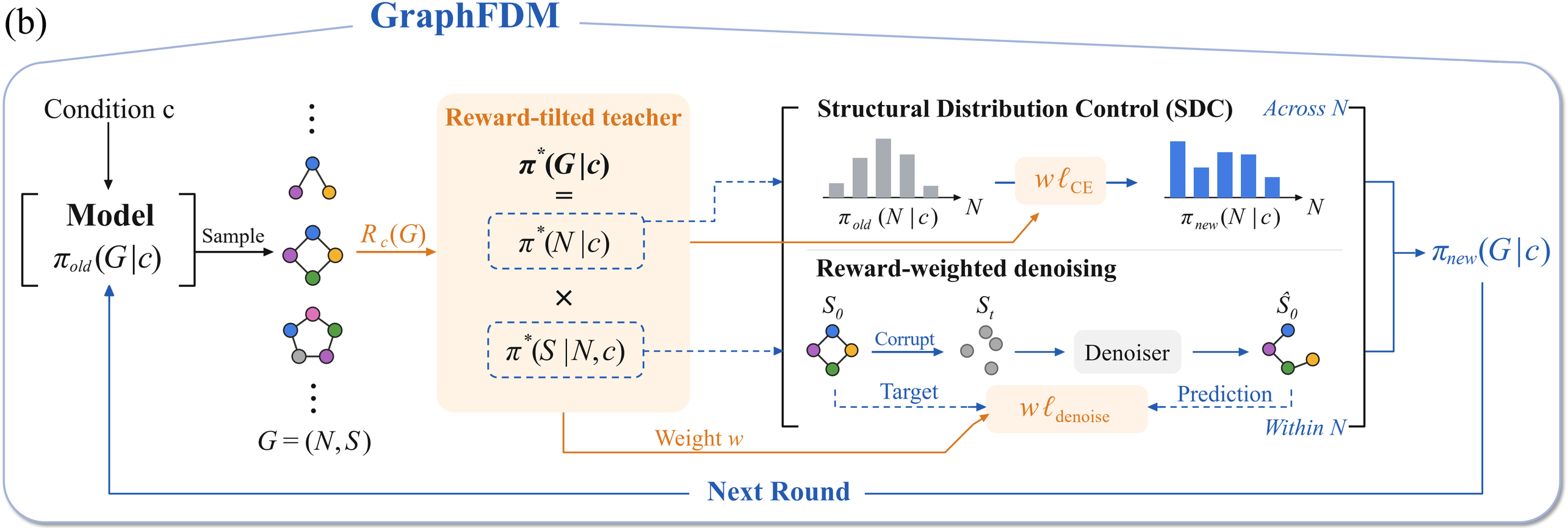
(a)


Reverse-policy RL (GRPO)
Generated Samples
Reward
w
w
w
Error Reinforcement
Model
Sample
Update
Valid
Targets
w1
w2
w3
Supervision

(b)


GraphFDM
Condition c
Model
$\pi_{old}(G|c)$
Sample
$G=(N,S)$
$R_c(G)$
Reward-tilted teacher
$\pi^*(G|c)$
=
$\pi^*(N|c)$
×
$\pi^*(S|N,c)$
Weight w
Structural Distribution Control (SDC)
Across N
$\pi_{old}(N|c)$
N
$w\ell_{CE}$
$\pi_{new}(N|c)$
N
Reward-weighted denoising
S_0
Corrupt
S_t
Denoiser
$\hat{S}_0$
Target
$w\ell_{denoise}$
Prediction
Within N
$\pi_{new}(G|c)$
Next Round